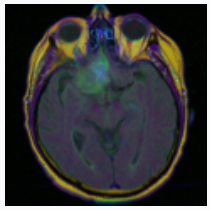
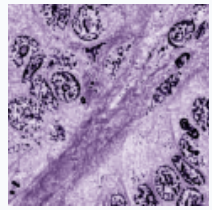
Stage 1: Multi-Modal Clinical Cohorts & Preprocessing


TCGA (MRI)


PANDA (Biopsy)


SIIM (X-Ray)


PanNuke (Tissue)

Patient-Isolated GroupKFold | **Macenko Stain Normalization**

Input $\mathbf{x} \in \mathbb{R}^{H \times W \times C}$

Stage 2: Shared Backbone Feature Extraction (VGG16 / MobileNetV2)

Block 1 128^2 → **Block 2** 64^2 → **Block 3** 32^2 → **Block 4** 16^2 → **Bottleneck z** $\mathbf{z} \in \mathbb{R}^{h \times w \times d}$

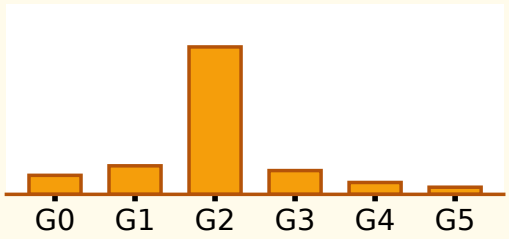
Lateral Skip Pathways $\{\mathbf{s}_l\}_{l=1}^{L}$

$\mathbf{z} \rightarrow$ **GAP**

$\mathbf{z} + \{\mathbf{s}_l\} \rightarrow \mathrm{Dec}_\phi$

Task 1: Classification Head

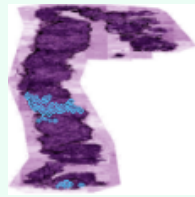
$\mathrm{GAP} \rightarrow \mathrm{MLP} \rightarrow \hat{\mathbf{y}}_{cls} \in \Delta^K$



Loss $\mathcal{L}_{cls}$ **(Weighted CCE)**

Task 2: Segmentation Head

$\mathrm{Dec}_\phi(\mathbf{z}, \{\mathbf{s}_l\}) \rightarrow \mathbf{I}_{seg} \in \mathbb{R}^{H \times W \times M}$



Loss $\mathcal{L}_{seg}$ **(Mask CE/BCE)**

Stage 4: Joint Optimization & Metric Disaggregation

$\mathcal{L}_{total} = w_{cls}(t)\,\mathcal{L}_{cls} + w_{seg}(t)\,\mathcal{L}_{seg}$ (Static or Adaptive Balancing)

Adaptive GradNorm: $\mathcal{L}_{grad}(t) = \sum_i \left| G_W^{(i)}(t) - \bar{G}_W(t)[r_i(t)]^\alpha \right| \rightarrow (w_{cls}, w_{seg})$

Disaggregated Metric: $\overline{\mathrm{Dice}}_{\mathrm{all}} = \rho \cdot q_{\mathrm{empty}} + (1-\rho) \cdot \overline{\mathrm{Dice}}_{\mathrm{fg}}$

Isolates Foreground Lesion Overlap ($\overline{\mathrm{Dice}}_{\mathrm{fg}}$) from Negative-Case Floor (ρ)